



BI VS BIG DATA – ROLE PLAY



Submitted By:

Name : Pratheek R S

USN : 1AY23CD048

Class &Section : 6A CSE(DS)

Course: Big Data Analytics

From Traditional BI to Big Data:

Introduction:

In this role play, a Business Manager believes that traditional Business Intelligence tools such as Excel, SQL databases, and centralized data warehouses are sufficient for organizational analytics. A Data Consultant, however, explains why modern enterprises must adopt Big Data technologies to manage rapidly growing, high-volume, high-velocity, and diverse data environments.

Conversation:

Business Manager:

I believe our Excel reports and SQL dashboards are sufficient for managing business analytics.

Data Consultant:

They are effective for structured data, but modern organizations generate much larger and more diverse datasets.

Business Manager:

What limitations do you see in traditional BI systems?

Data Consultant:

Traditional BI mainly handles structured data and works through batch processing, which limits real-time decision-making.

Business Manager:

But our data warehouse stores years of information efficiently.

Data Consultant:

Yes, but as data volume and velocity increase, centralized systems struggle to scale effectively.

Business Manager:

So how did Big Data evolve from traditional BI?

Data Consultant:

With the rise of social media, IoT devices, and online platforms, data began growing in volume, variety, and speed, requiring new technologies.

Business Manager:

What technologies are used to manage this scale?

Data Consultant:

Hadoop is used for distributed storage and parallel processing of massive datasets.

Business Manager:

And what about different data formats like JSON logs and customer feedback?

Data Consultant:

That is where NoSQL databases are useful, as they support flexible schemas and handle semi-structured and unstructured data.

Business Manager:

What business advantages does Big Data provide?

Data Consultant:

It enables real-time analytics, predictive insights, cost optimization, and improved customer experiences.

Business Manager:

So Big Data enhances traditional BI rather than replacing it?

Conclusion:

Traditional BI supports structured reporting but struggles with large and fast-growing data. Big Data technologies like Hadoop and NoSQL provide scalability and real-time analytics, helping businesses make better decisions.